EXAMENSARBETE INOM ELEKTROTEKNIK,
AVANCERAD NIVÅ, 30 HP
STOCKHOLM, SVERIGE 2020

Optimal Control Model for an Autonomous Underwater Vehicle

GUSTAV HOLM

KTH
SKOLAN FÖR ELEKTROTEKNIK OCH DATAVETENSKAP

Abstract

The main goal with this thesis was to develop an optimal control model for an autonomous underwater vehicle (AUV) and evaluate whether reference tracking using Model Predictive Control (MPC) based on a linear dynamics model describing all six degrees of freedom (DOF) is a suitable method for waypoint navigation. MPC is an advanced receding horizon optimal control method capable of including constraints in the optimization. The drawback with this method is that it is computationally heavy and since the dynamic AUV behavior is both complex and nonlinear there are two open research questions: Does MPC improve reference tracking and is it computationally feasible?

Optimal control requires accurate dynamics models of the targeted system to function and ensure robustness. At the start of this thesis the targeted AUV was an unmodelled vehicle and therefore modeling, model analysis and linearization techniques constitutes a big part of this thesis and creating a nonlinear dynamics model was a big task in this project.

To ensure that reference tracking using optimal control strategies was feasible, state-feedback solutions together with standard techniques for stability analysis, controllability and observability were investigated. A linear quadratic regulator (LQR) was designed using a linear time-invariant (LTI) dynamics model augmented with integral action and error dynamics to create a performance reference for the MPC implementation. The final step during this thesis was to implement MPC reference tracking using integral action on both states and inputs and the results from this controller was then compared to the results from the LQR controller in terms of performance.

A model generator capable of creating 6 DOF nonlinear dynamics models with varying complexity has been designed. This generator was used to derive two different linear models describing the dynamic AUV behavior. These linear models were analyzed and the necessary model parameters were estimated using simulation and then compared to physical data and live testing. Results from the LQR controller using an augmented error dynamics model is promising in terms of reference tracking. Due to a time varying reference and an under-actuated vehicle the resulting performance of the MPC controller does not match the results from the LQR controller. Overall, model-based optimal control show potential as a method for dynamic control and waypoint navigation for AUV applications, but further work with parameter estimation, model validation and control objective definition is required to ensure feasibility in a physical implementation.

Sammanfattning

Huvudmålet med detta examensarbete var att utveckla en modell för optimal styrning av en autonom undervattensfarkost och utvärdera om reglering med model predictive control (MPC) tillsammans med en linjär systemmodell som beskriver farkostens rörelse i alla sex frihetsgrader (DOF) är en lämplig metod för waypoint navigering. MPC är en avancerad metod för optimal styrning som använder en rörlig horisont och möjligör att begränsningar på tillstånd och styrsignaler kan inkluderas i optimeringen. Nackdelen med denna metod är att den är beräkningstung och eftersom farkostens dynamiska beteende är både komplext och icke-linjärt så är de två huvudfrågorna om en linjär systemmodell resulterar i tillräcklig hög prestanda och om beräkningstiden är tillräckligt låg.

Optimala regulatorer behöver noggranna dynamiska modeller för att fungera och säkerställa robusta lösningar. När detta examensarbete startade så fanns ingen dynamisk modell tillgänglig och därför utgjorde modellering, modellanalys och linjärisering en stor del av arbetet och skapandet av en ickelinjär systemmodell betraktades som en av huvuduppgifterna.

För att säkerställa att reglering med optimala styrmetoder var möjligt så undersöktes först tillståndsåterkoppling tillsammans med standardmetoder för stabilitetsanalys, styrbarhet och observerbarhet. En linjärkvadratisk regulator (LQR) skapades med en linjär dynamisk modell som förstärkts med felreglering och integrerade tillstånd för att användas som referens till MPC implementationen. Som sista steg skapades en MPC regulator förstärkt med integrerade tillstånd och styrsignaler. Därefter jämfördes resulaten från denna implementation med resultaten från LQR.

En modellgenerator kapabel till att skapa ickelinjära dynamiska modeller i 6 frihetsgrader designades. Den här generatorn användes sedan för att bygga två linjära modeller som beskriver farkostens dynamiska beteende. Dessa linjära modeller undersöktes och alla nödvändiga parametrar estimerades med hjälp av simulering och utvärderades sedan mot insamlad data och farkostens beteende vid fysiska tester. Resultaten från LQR med en modell för feldynamik är lovande med avseende på reglering. På grund av en tidsvarierande referens och att farkosten är underaktuerad är MPC regulatorns prestanda ej likvärdig med prestandan från LQR. Övergripande så visar modellbaserad optimal styrning stor potential som metod för dynamisk styrning och tillämpningar på autonoma undervattensfarkoster men mer arbete inom parameterestimering, modellanalys och formuleringen av styrobjektiv är nödvändigt för att kunna säkerställa prestanda och implementationsmöjligheterna i ett fysiskt system.

Acknowledgements

First of all I want to thank Jakob Kuttenkeuler for giving me the opportunity to write this thesis at the KTH Centre for Naval Architecture and for inviting me to be a part of the LoLo development team. To be a part of this team has provided both knowledge and understanding about maritime robotics and the craft itself which has been a valuable resource throughout this thesis.

I am grateful for the help and support provided by my supervisor Clemens Deutsch who, throughout this thesis, has pushed me in the right direction and helped me to put focus and effort into reaching as far as possible.

I extend my gratitude to Jonas Mårtensson at the Division of Decision and Control Systems for taking on the role of examiner and Joana Filipa Gouveia Fonseca for helping me with control related problems and for providing advice on implementation aspects of MPC and LQR.

Lastly I want to thank my family and friends for their support, even though not always understanding the theoretical aspects of my work.

Stockholm, December 2019
Gustav Holm

Contents

Definitions

LoLo	-	*Name of the vehicle*
MPC	-	Model Predictive Control
LTV	-	Linear Time-Varying
LTI	-	Linear Time-Invariant
DOF	-	Degree of Freedom
AUV	-	Autonomous Underwater Vehicle
LQR	-	Linear Quadratic Regulation
KTH	-	*Royal Institute of Technology*
SMaRC	-	*Swedish Maritime Robotics Centre*
SISO	-	Single-Input Single-Output
MIMO	-	Multiple-Input Multiple-Output
SF	-	State-Feedback
NED	-	North East Down
BODY	-	Body
VP	-	Vessel Parallel
SNAME	-	*Society of Naval Architects and Marine Engineers*
VBS	-	Variable Buoyancy System
WGS	-	*World Geodetic System*
BIBO	-	Bounded-Input Bounded-Output
ARE	-	Algebraic Riccati Equation
RPM	-	Revolutions Per Minute
ZOH	-	Zero Order Hold
AOA	-	Angle of Attack

Notations

CG	-	Center of gravity
CB	-	Center of buoyancy
CO	-	Center of arbitrary coordinate system
l	-	Length of vehicle
b	-	Width of vehicle
h	-	Height of vehicle
x_g	-	Distance from CO to CG {x}
y_g	-	Distance from CO to CG {y}
z_g	-	Distance from CO to CG {z}
x_b	-	Distance from CO to CG {x}
y_b	-	Distance from CO to CG {y}
z_b	-	Distance from CO to CG {z}
A_{yz}	-	Front cross section area
A_{xy}	-	Top cross section area
A_{xz}	-	Side cross section area
A_r	-	Area of rudders
A_e	-	Area of elevator
A_ev	-	Area of elevons
l_{roll}	-	Distance from rudder to CO {z}
l_{pitch}	-	Distance from elevator to CO{x}
l_{yaw}	-	Distance from rudder to CO {x}
l_{px}	-	Distance from propeller to CO {x}
l_{py}	-	Distance from propeller to CO {y}
l_{epitch}	-	Distance from elevons to CO {x}
l_{eroll}	-	Distance from elevons to CO {y}
m	-	Craft weight
I_x	-	Inertia in roll
I_y	-	Inertia in pitch
I_z	-	Inertia in yaw
I_{xy}	-	Coupled inertia for roll and pitch
I_{xz}	-	Coupled inertia for roll and yaw
I_{yz}	-	Coupled inertia for pitch and yaw
ρ	-	Density of water
g	-	Gravitational acceleration
W	-	Gravitation force
B	-	Buoyancy force
$X_{\dot{u}}$	-	Added mass in surge
$X_{\dot{v}}$	-	Added mass in surge due to sway
$X_{\dot{w}}$	-	Added mass in surge due to heave
$X_{\dot{p}}$	-	Added mass in surge due to roll
$X_{\dot{q}}$	-	Added mass in surge due to pitch
$X_{\dot{r}}$	-	Added mass in surge due to yaw
$Y_{\dot{v}}$	-	Added mass in sway

$Y_{\dot{w}}$	-	Added mass in sway due to heave
$Y_{\dot{p}}$	-	Added mass in sway due to roll
$Y_{\dot{q}}$	-	Added mass in sway due to pitch
$Y_{\dot{r}}$	-	Added mass in sway due to yaw
$Z_{\dot{w}}$	-	Added mass in heave
$Z_{\dot{p}}$	-	Added mass in heave due to roll
$Z_{\dot{q}}$	-	Added mass in heave due to pitch
$Z_{\dot{r}}$	-	Added mass in heave due to yaw
$K_{\dot{p}}$	-	Added mass in roll
$K_{\dot{q}}$	-	Added mass in roll due to pitch
$K_{\dot{r}}$	-	Added mass in roll due to yaw
$M_{\dot{q}}$	-	Added mass in pitch
$M_{\dot{r}}$	-	Added mass in pitch due to yaw
$N_{\dot{r}}$	-	Added mass in yaw
X_u	-	Linear drag coefficient in surge
$X_{\|u\|u}$	-	Quadratic drag coefficient in surge
Y_v	-	Linear drag coefficient in sway
$Y_{\|v\|v}$	-	Quadratic drag coefficient in sway
Z_w	-	Linear drag coefficient in heave
$Z_{\|w\|w}$	-	Quadratic drag coefficient in heave
K_p	-	Linear drag coefficient in roll
$K_{\|p\|p}$	-	Quadratic drag coefficient in roll
M_q	-	Linear drag coefficient in pitch
$M_{\|q\|q}$	-	Quadratic drag coefficient in pitch
N_r	-	Linear drag coefficient in yaw
$N_{\|r\|r}$	-	Quadratic drag coefficient in yaw
T	-	Thrust generated by propeller
L	-	Lift force generated by actuator
D	-	Drag force generated by actuator
δ	-	Rudder deflection angle
α	-	Vehicle angle of attack
R	-	Engine RPM
t	-	Time
h	-	Sampling interval
C_d	-	Drag coefficient
C_L	-	Lift coefficient

1 Introduction

In modern naval architecture the focus on robotics, automation and smart solutions for maritime research are continuously increasing. Development of maritime platforms capable of autonomously carrying out research missions is a difficult process that combines research and engineering within many different fields. The Royal Institute of Technology's (KTH) Centre for Naval Architecture together with the Swedish Maritime Robotics Centre (SMaRC) is currently working on an experimental autonomous underwater vehicle named LoLo that will serve as a demonstration platform for new technologies and equipment [1]. This vehicle is designed for long range autonomous underwater research missions and is capable of carrying different types of equipment and payloads.

1.1 Background

The vehicle's ability to perform efficient and precise maneuvers during these missions can be directly linked to mission success and will increase maneuverability, reliability and extend mission duration. To achieve this, an advanced control system is required and this in turn requires an accurate dynamics model of the vehicle.

1.2 Purpose

There are several reasons why a more advanced control system would be beneficial for this vehicle. The complex couplings in the vehicle's dynamics requires either multiple-input multiple-output (MIMO) control or simplifications that are not physically accurate but enables the use of single-input single-output (SISO) control systems. Working with simplifications and traditional SISO control methods is an effective technique that often yields acceptable results. The drawback is that simplifications and assumptions often reduce or restrict controller performance, which in this case limits maneuvering capabilities or performance. Another thing to consider is that each SISO controller only have one channel, which means that to fully control the vehicle several controllers are needed. These controllers may have contradicting objectives and make for a more advanced software structure. A MIMO control system using traditional Proportional-Integral-Derivative (PID) control have the benefit of combining all the controllers into one single control system. The drawback here is that dealing with performance in traditional MIMO control can be hard and trade-off between different objectives will be introduced. Designing a state-feedback (SF) solution that operates in the time domain results in several benefits. All the existing controllers can be replaced by one single control system and advanced methods involving optimization can be implemented to derive optimal solutions.

1.3 Objective

The main objective of this thesis was to evaluate whether model-predictive control (MPC) reference tracking using a linearized dynamics model of the vehicle LoLo is a suitable method for waypoint tracking or regulation in terms of performance and computational time. To reach this final objective, there are several other tasks or goals that need to be achieved.

1.4 Method

This thesis was divided into three phases.

- Phase 1: The first phase covers all the theoretical aspects and includes a four week long study in naval architecture, modeling and control of naval vessels. The purpose of this phase was to familiarize with the vehicle's existing systems, prerequisites and emphasis was put on how modeling and control was applied in practice by naval architects and marine engineers.

- Phase 2: The second phase included all the modeling. Here, the purpose was to derive a complete dynamics model describing all 6 degrees of freedrom (DOF) of the vehicle. From there the dynamics model was translated into a state-space representation and parameters describing dynamic behavior was estimated. The final step in phase two was to create the linear models representing the vehicle's nonlinear dynamics.

- Phase 3: The third phase covered control theory and methods. In this phase, the final goal was to derive the described MPC controller. In order to reach this implementation several other steps were taken. The dynamics model was first evaluated in terms of stability, controllability and observability. After this, the model was augmented with necessary components to enable regulation and tracking. From here, standard state-feedback using pole placement was investigated to make sure that a state-feedback solutions could be derived and that the problem was feasible. Lastly, one LQR controller and one MPC controller were derived, and their performance was evaluated.

1.5 Limitations

At the start of this thesis LoLo was an unmodelled vehicle. Since advanced control techniques such as LQR and MPC relies heavily on the accuracy of dynamics models this means that a lot of time has to be spent on modeling and model evaluation which will be a big part of this thesis.

Vehicle symmetry was another important aspect during the modeling phase. LoLo is a slender vehicle, which means that it is long in relation to its height and width. This in combination with left-right symmetry and nearly bottom-top symmetry allows for a number of simplifications. These simplifications drastically reduce the number of elements in the matrices that describe vehicle dynamics. This will be further explained in Section 3.

When modeling the vehicle LoLo, a modeling technique using hydrodynamic derivatives was used. These hydrodynamic derivatives are parameters that describe how different forces and torques affect the vehicle and are strictly design-dependent. Determining values for each of these parameters is time consuming and a big project by itself. Therefore, dummy values, simplifications and assumptions will instead be used during this thesis. The result of this is that the model will not be a perfect resemblance of the physical system but sufficient to evaluate performance of different controllers and determine feasibility for advanced control methods.

Even though the vehicle dynamics are more accurately described by a nonlinear model on the form $\dot{\boldsymbol{x}} = \boldsymbol{f}(t, \boldsymbol{\nu}, \boldsymbol{\tau})$, the intention with this thesis is to evaluate whether MPC using linear dynamics can be implemented to generate good reference tracking performance. Using a linear model to describe the dynamics of a nonlinear system have both pros and cons. As mentioned before, the accuracy is of great concern. If the model dynamics are too far from the actual vehicle dynamics the resulting controllers will not perform well enough. However, the upside of using linear models is that the computational capacity required to derive the control solutions is drastically reduced which might be crucial for advanced methods like MPC to be feasible in practice. This creates a trade-off between model accuracy and computational effort, and the question stands if a sufficiently accurate linear model can be derived and whether the computational time needed by the MPC controller using this linear model is acceptable.

The vehicle has two types of actuators, control surfaces and propellers. The control surfaces are represented by rudders, elevator and elevons and constitute the majority of the actuators. In this thesis control surfaces were modeled as symmetrical foils and when deflected an angle δ under forward speed u they creates two forces, commonly described as lift (L) and drag (D). This means that, depending on the rudder position, each rudder has the capability of producing a maximum of two forces and two moments. All of these forces and moments will be described in Section 3, but since $\frac{L}{D} \approx 5$ for low values of δ the effect of drag was neglected in the controllers. Some of the moments induced by lift and drag were also neglected due to short distances between CO and the actuator.

1.6 Software

During this thesis, Matlab was the main software used. Matlab is an advanced computing environment that uses it's own programming language and a wide variety of built in toolboxes that simplify calculations, for example the symbolic math toolbox that enables solving and manipulation of symbolic valued math functions [2] that has been used extensively throughout this thesis. For constrained optimization and MPC, the third-party toolbox YALMIP [3] has been used. YALMIP is an optimization toolbox that provides a framework which enables the user to formulate constrained optimization problems with an alternative setup and a variety of different solvers. For optimization, Matlab's built-in solvers Quadprog and Fmincon have been used together with the external solver Gurobi [4]. All of these solvers have their own strengths and weaknesses, but overall the more advanced Gurobi solver has yielded the best results in terms of computation time.

1.7 Thesis Outline

This report starts with Section 2 about the theoretical concepts necessary to understand the methods used in this thesis. Section 3 describes the modeling phase of this thesis. There, the techniques used to derive a complete linear dynamics model from a physical system is presented. In Section 4 the concepts used to analyze and augment the dynamics models together with the theoretical aspects of state-feedback, LQR and MPC are presented. A short description of the vehicle's existing systems is also included. Section 5 covers the results obtained during this thesis from both modeling and control. In Section 6 all the results and key concepts are evaluated. Lastly, Section 7 and 8 presents some final thoughts about the method used and future work needed to further improve the dynamics model and controllers derived during this thesis.

2 Theoretical Concepts

In this section, some of the key concepts necessary to understand the work performed during this thesis are described together with a description of the vehicle LoLo.

2.1 SNAME notation of motion

Throughout this thesis and in this report the Society of Naval Architects and Marine Engineers (SNAME) notation of motion [5] was used to describe forces, moments, positions, angles, linear and angular velocities. This notation facilitates keeping track of different quantities, which is of great importance when working with systems with complex and coupled dynamics and simplifies both calculation and analysis. In Table 1 an overview of the notation is presented and in Figure 1 the DOFs are illustrated from the vehicle perspective.

Table 1: SNAME notation for marine vessels

DOF		Forces and moments	Linear and angular velocities	Positions and euler angles
1	Motion in x (Surge)	X	u	x
2	Motion in y (Sway)	Y	v	y
3	Motion in z (Heave)	Z	w	z
4	Rotation around x (Roll)	K	p	ϕ
5	Rotation around y (Pitch)	M	q	θ
6	Rotation around z (Yaw)	N	r	ψ

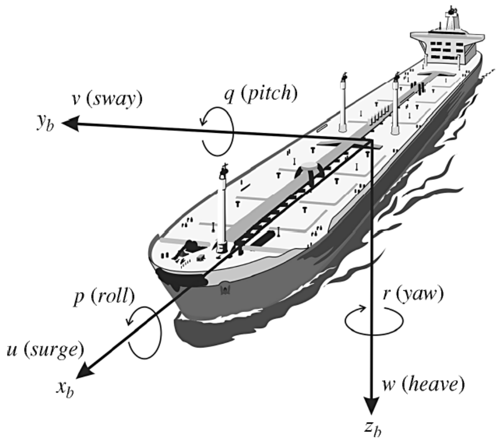

Figure 1: Motions and rotations in a vehicle perspective [6]

2.2 Degrees of Freedom

The number of DOFs used in the equations of motion is important, because it defines how many differential equations are needed to fully describe the vehicle's position, displacement and orientation in a state-space model. A vehicle operating freely in a three dimensional (3D) space can be described by a maximum of 6 DOFs, three translational and three rotational components, which results in a state-space model of order 12. Depending on the application, the number of DOFs can be reduced even when working in a 3D environment, but doing this for a vehicle with complex and coupled dynamics often requires heavy simplifications. DOFs are also relevant for the concept of actuation in vehicles. For a vehicle to be fully actuated it requires independent actuation in each DOF. If this is not satisfied, the vehicle is referred to as underactuated and consequently only control objectives with the same number of DOFs as there are independent actuators can be solved.

2.3 Vehicle Description

LoLo is an AUV with dimensions according to Table 2. The vehicle is equipped with two electrical motors with attached propellers, two coupled rudders, two elevons and one elevator. Together, these 7 actuators are capable of producing forces and moments in all 6 DOFs. Since all actuators except the electrical motors are control surfaces, the vehicle's ability to perform certain maneuvers become highly speed dependent. This, in combination with the fact that rudders produce both moments and forces result in weak actuation in both sway and heave, which in turn leads to the vehicle being underactuated. Even though a variable buoyancy system (VBS) has not been implemented, the AUV is assumed to be neutrally buoyant to simplify modeling and to remove some of the speed dependency. The AUV operates at relative low speeds of $1-3$ m/s, the engines are limited to ± 500 $[RPM]$ and rudders operate in between ± 20 $degrees$. CG and CB are located at the same x and y coordinates, whereas the distance between z_g and z_b is 0.4 m. Further, the vehicle's moment of inertia are calculated assuming a block shape body and the vehicle design is presented in Figure 2.

Figure 2: Vehicle design

Table 2: LoLo's physical data

l	-	3.8 m
b	-	1.05 m
h	-	0.6 m
m	-	300 kg
I_x	-	37 kgm^2
I_y	-	370 kgm^2
I_z	-	389 kgm^2

2.4 Coordinate Frames

Throughout this thesis a number of different coordinate frames have been used. Understanding what different notations mean, how vehicle dynamics are described and how to transform different motions between coordinate frames was an important aspect when both modeling and when regulating the system to a desired target or way point. Below is a short description of the three most important coordinate frames, North-East-Down (NED), Body Coordinates (BODY) and Vessel Parallel Coordinates (VP). The center of origin in each coordinate system can be chosen arbitrarily, e.g. the fore of the vehicle to create a safety margin for movement in surge, but to simplify modeling, the best location is usually when it coincides with CG and CB in the lateral coordinates x and y.

2.4.1 NED {n}

North East Down (NED) coordinates, denoted by $\{n\} = (x_n, y_n, z_n)$ is a vehicle-fixed coordinate frame defined in relation to the Earth's reference ellipsoid [6]. The $x-axis$ in this coordinate frame points toward north and the $y-axis$ points toward east as illustrated in Figure 3. This is the coordinate frame commonly used in everyday life known as latitude and longitude or flat-earth navigation and the $z-axis$ points directly downwards. In this thesis the following notation was used for lateral and angular positions in the NED frame:

$$\boldsymbol{\eta} = [N = x_n, E = y_n, D = z_n, \phi, \theta, \psi]$$

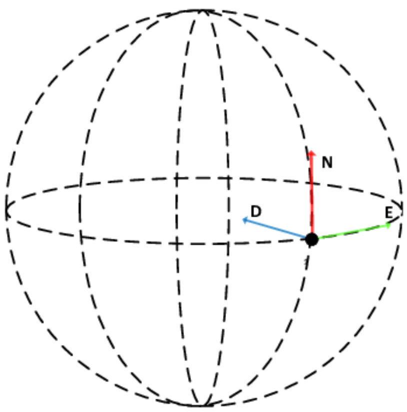

Figure 3: NED coordinate frame

2.4.2 BODY {b}

Body (BODY) coordinates, denoted by $\{b\} = (x_b, y_b, z_b)$ is a vehicle-fixed local coordinate frame, where position and orientation are described relative to the inertial frame (NED). Linear and angular velocities are instead expressed in a body-fixed frame where x_b stretches from aft to fore, y_b is directed to starboard and z_b stretches from top to bottom as illustrated in Figure 4. In this thesis the following notation was used for velocities in the BODY frame:

$$\boldsymbol{\nu} = [u, v, w, p, q, r] \tag{1}$$

Integrating Equation 1 with respect to time gives the lateral and angular positions in the BODY frame:

$$\frac{\partial \boldsymbol{\nu}}{\partial t} = \boldsymbol{x} = [x_b, y_b, z_b, \phi, \theta, \psi]$$

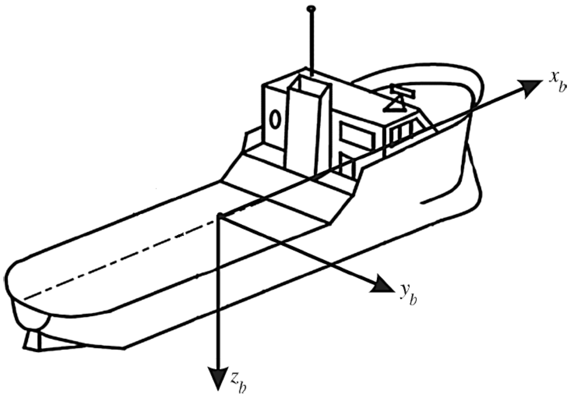

Figure 4: BODY coordinate frame [6]

2.4.3 VP {p}

Vessel parallel (VP) coordinates is a coordinate frame used to linearize the equations of motion under certain assumptions [6]. Simplified, VP coordinates can be seen as a mix between the NED and the BODY frames translated into a single reference frame. A description of how this works and how to derive this reference frame will be presented in Section 3.

2.5 Positions and Euler Angles

In this thesis Euler angles were used to describe the relative positions and rotations between the NED, BODY and VP coordinate frames. The reason for this is simply that Euler angles are more intuitive and easier to work with than quaternions and since the nonlinear equations of motion are later linearized the difference is insignificant. Moreover, the vehicle is not designed to perform any advanced maneuvers such as rolls or loops which means that disadvantages of using Euler angles such as Gibal locks [7] will not occur. The nonlinear relationship between NED and BODY is described by:

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J}_{\Theta}(\boldsymbol{\eta})\boldsymbol{\nu} \tag{2}$$

where

$$\boldsymbol{J}_{\Theta}(\boldsymbol{\eta}) = \begin{bmatrix} \boldsymbol{R}(\boldsymbol{\Theta}) & \boldsymbol{0_{3x3}} \\ \boldsymbol{0_{3x3}} & \boldsymbol{T}_{\Theta}(\boldsymbol{\Theta}) \end{bmatrix} \iff \boldsymbol{J}_{\Theta}^{-1}(\boldsymbol{\eta}) = \begin{bmatrix} \boldsymbol{R}(\boldsymbol{\Theta})^{\boldsymbol{T}} & \boldsymbol{0_{3x3}} \\ \boldsymbol{0_{3x3}} & \boldsymbol{T}_{\Theta}^{-1}(\boldsymbol{\Theta}) \end{bmatrix}$$

and

$$\boldsymbol{R}(\boldsymbol{\Theta}) = \begin{bmatrix} \cos\psi\cos\theta & -\sin\psi\cos\phi+\cos\psi\sin\theta\sin\phi & \sin\psi\sin\phi+\cos\psi\cos\phi\sin\theta \\ \sin\psi\cos\theta & \cos\psi\cos\phi+\sin\phi\sin\theta\sin\psi & -\cos\psi\sin\phi+\sin\theta\sin\psi\cos\phi \\ -\sin\theta & \cos\theta\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

and

$$T_\Theta(\Theta) = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \frac{\sin\phi}{\cos\theta} & \frac{\cos\phi}{\cos\theta} \end{bmatrix}$$

2.6 Hydrodynamic Derivatives

Hydrodynamic derivatives are widely used to model the hydrodynamic forces of dynamical systems such as aircrafts and naval vessels. This concept assumes that added mass due to viscous effects and damping which is the fluid resistance can be expressed by constant parameters when the vehicle is moving in relative calm water with none to low wave excitation. As an example, the linear hydrodynamic damping in X due to a velocity in surge u would be expressed as $X = X_u u$ and the simplification can be seen in Equation 3.

$$\boxed{X = \frac{1}{2}\rho C_d A_{yz} u^2 \approx X_u u} \tag{3}$$

2.7 Vehicle Symmetry

The downside of modeling with hydrodynamic derivatives is that the value for a large number of parameters need to be determined which can be both difficult and time consuming. Luckily, symmetry conditions can be used to significantly reduce the number of parameters needed. $\boldsymbol{M}$ is generally a full matrix describing the vehicles mass and moment of inertia, but since $\boldsymbol{M} = \boldsymbol{M^t} > 0$ and the vehicle is assumed to be both bottom-top symmetrical and port-starboard symmetrical the following cases can be derived.

- XY-symmetry (bottom-top) results in the following case:

$$\boldsymbol{M} = \begin{bmatrix} m_{11} & m_{12} & 0 & 0 & 0 & m_{16} \\ m_{21} & m_{22} & 0 & 0 & 0 & m_{26} \\ 0 & 0 & m_{33} & m_{34} & m_{35} & 0 \\ 0 & 0 & m_{43} & m_{44} & m_{45} & 0 \\ 0 & 0 & m_{53} & m_{54} & m_{55} & 0 \\ m_{61} & m_{62} & 0 & 0 & 0 & m_{66} \end{bmatrix}$$

- XZ-symmetry (port-starboard) results in the following case:

$$\boldsymbol{M} = \begin{bmatrix} m_{11} & 0 & m_{13} & 0 & m_{15} & 0 \\ 0 & m_{22} & 0 & m_{24} & 0 & m_{26} \\ m_{31} & 0 & m_{33} & 0 & m_{35} & 0 \\ 0 & m_{42} & 0 & m_{44} & 0 & m_{46} \\ m_{51} & 0 & m_{53} & 0 & m_{55} & 0 \\ 0 & m_{62} & 0 & m_{64} & 0 & m_{66} \end{bmatrix}$$

- Combining the two previous cases results in:

$$\boldsymbol{M} = \begin{bmatrix} m_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & m_{22} & 0 & 0 & 0 & m_{26} \\ 0 & 0 & m_{33} & 0 & m_{35} & 0 \\ 0 & 0 & 0 & m_{44} & 0 & 0 \\ 0 & 0 & m_{53} & 0 & m_{55} & 0 \\ 0 & m_{62} & 0 & 0 & 0 & m_{66} \end{bmatrix} \tag{4}$$

3 Modeling

The second phase of this thesis covered modeling of the AUV. Initially the goal was to create a complete nonlinear dynamics model of the vehicle using the general technique described by Fossen's robot like vectorial model [6] and then customize the model to give a good representation of the target vehicle LoLo and its dynamics. When the model was created the equations of motion needed to be converted to state-space form. From there different linearization techniques were investigated and in the final step the model was discretized. Initially the plan was to place the coordinate frame origin CO in the bow of the vehicle, but this was later changed and the origin was placed in CG to simplify both the equations of motion and the linearization process. The created model has the form:

$$\boxed{\begin{aligned}\dot{\boldsymbol{\eta}} &= \boldsymbol{J}_{\Theta}(\boldsymbol{\eta})\boldsymbol{\nu} && (5)\\ \boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{g}(\boldsymbol{\eta}) + \boldsymbol{g_0} &= \boldsymbol{\tau} && (6)\\ \boldsymbol{\eta} &= [N, E, D, \phi, \theta, \psi] && (7)\\ \boldsymbol{\nu} &= [u, v, w, p, q, r] && (8)\end{aligned}}$$

where the positions and angles ($\boldsymbol{\eta}$) are described in NED and the linear and angular velocities ($\boldsymbol{\nu}$) are described in BODY. $\boldsymbol{M}$ represent the vehicle's mass and inertia, $\boldsymbol{C}(\boldsymbol{\nu})$ represent Coriolis and centripetal forces, $\boldsymbol{D}(\boldsymbol{\nu})$ represent damping forces, $\boldsymbol{\tau}$ represent actuator forces and $\boldsymbol{g}(\boldsymbol{\eta}) + \boldsymbol{g_0}$ represent restoring forces. Section 3.1 to 3.5 describes how each of these matrices are derived.

3.1 Mass and Inertia

The mass and inertia matrix $\boldsymbol{M}$ can be seen as a combination of three different matrices:

$$\boxed{\boldsymbol{M} = \boldsymbol{M}_{RB} + \boldsymbol{M}_T + \boldsymbol{M}_{AM}}$$

where $\boldsymbol{M}_{RB}$ is the rigid body mass matrix, $\boldsymbol{M}_T$ is a matrix describing mass and inertia changes due to reference frame position and orientation in relation to CG and $\boldsymbol{M}_{AM}$ is a full matrix of hydrodynamic derivatives describing added mass due to viscous effects. The full mass and inertia matrix has the form:

$$\boldsymbol{M} = \begin{bmatrix} m-X_{\dot{u}} & -X_{\dot{v}} & -X_{\dot{w}} & -X_{\dot{p}} & mz_g-X_{\dot{q}} & my_g-X_{\dot{r}} \\ -X_{\dot{v}} & m-Y_{\dot{v}} & -Y_{\dot{w}} & -mz_g-Y_{\dot{p}} & -Y_{\dot{q}} & mx_g-Y_{\dot{r}} \\ -X_{\dot{w}} & -Y_{\dot{w}} & m-Z_{\dot{w}} & my_g-Z_{\dot{p}} & -mx_g-Z_{\dot{q}} & -Z_{\dot{r}} \\ -X_{\dot{p}} & -mz_g-Y_{\dot{p}} & my_g-Z_{\dot{p}} & I_x-K_{\dot{p}} & -I_{xy}-K_{\dot{q}} & -I_{zx}-K_{\dot{r}} \\ mz_g-X_{\dot{q}} & -Y_{\dot{q}} & -mx_g-Z_{\dot{q}} & -I_{xy}-K_{\dot{q}} & I_y-M_{\dot{q}} & -I_{yz}-M_{\dot{r}} \\ -my_g-X_{\dot{r}} & mx_g-Y_{\dot{r}} & -Z_{\dot{r}} & -I_{zx}-K_{\dot{r}} & -I_{yz}-M_{\dot{r}} & I_z-N_{\dot{r}} \end{bmatrix}$$

Using the case described by Equation 4 together with a coordinate frame origin located in $CG(x_g, y_g, z_g = 0)$ and the assumption that $\boldsymbol{M}_{RB} >> \boldsymbol{M}_{AM}$, the mass and inertia matrix can be reduced to the following form:

$$
\boldsymbol{M} = \begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & 0 & 0 & I_x & 0 & 0 \\ 0 & 0 & 0 & 0 & I_y & 0 \\ 0 & 0 & 0 & 0 & 0 & I_z \end{bmatrix} \tag{9}
$$

3.2 Coriolis and Centripetal Forces

The Coriolis-centripetal matrix can be parameterized in several different ways. Since $\boldsymbol{C(\nu) = -C^T(\nu)}$ [8] the Coriolis-centripetal matrix can be derived from the system mass and inertia matrix described in Equation 9 in the following way.

Let

$$
\boldsymbol{M} = \boldsymbol{M}^T = \begin{bmatrix} \boldsymbol{M_{11}} & \boldsymbol{M_{12}} \\ \boldsymbol{M_{21}} & \boldsymbol{M_{22}} \end{bmatrix}
$$

and

$$
\begin{aligned}
\boldsymbol{\nu_1} &= [u, v, w]^T \\
\boldsymbol{\nu_2} &= [p, q, r]^T
\end{aligned}
$$

A skew symmetrical 3x3 matrix can be derived according to:

$$
\boldsymbol{S}(\boldsymbol{\alpha}) = -\boldsymbol{S^T}(\boldsymbol{\alpha}) = \begin{bmatrix} 0 & -\alpha_3 & \alpha_2 \\ \alpha_3 & 0 & -\alpha_1 \\ -\alpha_2 & \alpha_1 & 0 \end{bmatrix}, \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}
$$

and the Coriolis-centripetal matrix was parameterized to:

$$
\boldsymbol{C}(\boldsymbol{\nu}) = \begin{bmatrix} \boldsymbol{0_{3x3}} & -\boldsymbol{S}(\boldsymbol{M_{11}\nu_1} + \boldsymbol{M_{12}\nu_2}) \\ -\boldsymbol{S}(\boldsymbol{M_{11}\nu_1} + \boldsymbol{M_{12}\nu_2}) & -\boldsymbol{S}(\boldsymbol{M_{21}\nu_1} + \boldsymbol{M_{22}\nu_2}) \end{bmatrix}
$$

By using the reduced mass and inertia matrix described by Equation 9 the resulting form on the Coriolis-centripetal matrix become:

$$
\boldsymbol{C}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & 0 & mw & -mv \\ 0 & 0 & 0 & -mw & 0 & mu \\ 0 & 0 & 0 & mv & -mu & 0 \\ 0 & mw & -mv & 0 & I_z r & -I_y q \\ -mw & 0 & mu & I_z r & 0 & I_x p \\ mv & -mu & 0 & I_y q & -I_x p & 0 \end{bmatrix} \tag{10}
$$

3.3 Damping Forces

In general the damping of a submerged AUV is both nonlinear and coupled [6]. To fully describe this dynamic behavior a matrix on the following form is required:

$$\boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \begin{bmatrix} |\boldsymbol{\nu}|^{\boldsymbol{T}}\boldsymbol{D_{n1}}\boldsymbol{\nu} \\ |\boldsymbol{\nu}|^{\boldsymbol{T}}\boldsymbol{D_{n2}}\boldsymbol{\nu} \\ |\boldsymbol{\nu}|^{\boldsymbol{T}}\boldsymbol{D_{n3}}\boldsymbol{\nu} \\ |\boldsymbol{\nu}|^{\boldsymbol{T}}\boldsymbol{D_{n4}}\boldsymbol{\nu} \\ |\boldsymbol{\nu}|^{\boldsymbol{T}}\boldsymbol{D_{n5}}\boldsymbol{\nu} \\ |\boldsymbol{\nu}|^{\boldsymbol{T}}\boldsymbol{D_{n6}}\boldsymbol{\nu} \end{bmatrix}$$

where $|\boldsymbol{\nu}| = [|u|, |v|, |w|, |p|, |q|, |r|]^T$ and $D_{ni}, i = 1, 2, 3, 4, 5, 6$ are full matrices with dimension 6x6. This description requires a large number of unknown parameter values and is therefore not practical. One method used to simplify the expression for damping is to separate it into two terms, one linear and one quadratic:

$$\boldsymbol{D}(\boldsymbol{\nu}) = [\boldsymbol{D} + \boldsymbol{D_N}(\boldsymbol{\nu})]$$

and then assume that the vehicle only performs non-coupled motions. This results in the following expression for damping:

$$\begin{aligned} \boldsymbol{D}(\boldsymbol{\nu}) = &-diag\{X_u, Y_v, Z_w, K_p, M_q, N_r\} \\ &- diag\{X_{|u|u}|u|, Y_{|v|v}|v|, Z_{|w|w}|w|, K_{|p|p}|p|, M_{|q|q}|q|, N_{|r|r}|r|\} \end{aligned} \tag{11}$$

3.4 Restoring Forces

Restoring forces include everything that force the vehicle back to an equilibrium where $(\phi, \theta = 0)$. For LoLo this includes two types of forces, displacement forces $(\boldsymbol{g}(\boldsymbol{\eta}))$ due to the location of CG and CB together with forces originating from ballast tank displacement $\boldsymbol{g_0}$. The ballast tank system, refereed to as VBS, is not yet implemented and the tank configuration is not determined. Therefore, $\boldsymbol{g_0} = 0$ but the expressions are still included in the report for future use. The full expression of restoring forces become:

$$\boldsymbol{g}(\boldsymbol{\eta}) = \begin{bmatrix} (W-B)\sin\theta \\ -(W-B)\cos\theta\sin\phi \\ -(W-B)\cos\theta\sin\phi \\ -(y_gW - y_bB)\cos\theta\cos\phi + (z_gW - z_bB)\cos\theta\sin\phi \\ (z_gW - z_bB)\sin\theta + (x_gW - x_bB)\cos\theta\cos\phi \\ -(x_gW - x_bB)\cos\theta\sin\phi - (y_gW - y_bB)\sin\theta \end{bmatrix} \tag{12}$$

The vehicle is assumed to be neutrally buoyant $(W = B)$ and placing the coordinate frame origin properly results in $x_g, x_b, y_g, y_b, z_g = 0$ and Equation 12 can be reduced to:

$$g(\boldsymbol{\eta}) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -z_b B \cos\theta \sin\phi \\ -z_b B \sin\theta \\ 0 \end{bmatrix} \tag{13}$$

Below is the expression for forces due to ballast tank use, where V_i is the volume of tank i.

$$\boldsymbol{g}_0 = \begin{bmatrix} 0 \\ 0 \\ -Z_{ballast} \\ -K_{ballast} \\ -M_{ballast} \\ 0 \end{bmatrix} = \rho g \begin{bmatrix} 0 \\ 0 \\ -\sum_{i=1}^{n} V_i \\ -\sum_{i=1}^{n} y_i V_i \\ \sum_{i=1}^{n} x_i V_i \\ 0 \end{bmatrix} = \mathbf{0} \tag{14}$$

3.5 Actuators and Control Surfaces

One of the most important aspects of modeling a vehicle is to accurately describe its actuators. To achieve a good physical description, both the magnitude of each force as well as how it enters or affects the system is required. In this thesis, all rudders and fins are modelled as symmetrical foils [9] and the thrust generated by the propellers are derived from provided propeller design data. In total, the vehicle has 7 actuators as described in Section 3.5 but the two rudders being mechanically coupled results in 6 possible control actions. Further more, the elevons attached to the sides of the vehicle were not implemented until mid November and therefore only included in the dynamics model and not the model used by the controllers. Equation 15 and 16 are the formulas used to determine the rudder, elevator and elevon forces. C_L is a lift coefficient and was approximated with 0.1δ according to typical data for symmetrical foils illustrated by Figure 5. The approximation used in Equation 16 is valid for the small control surface deflection angles assumed in this thesis.

$$L = \frac{1}{2} A \rho v^2 C_L \tag{15}$$

$$D \approx \sin\delta L \tag{16}$$

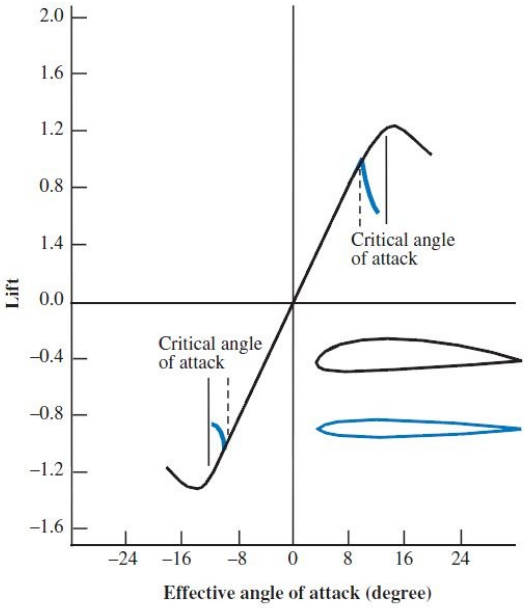

Figure 5: Lift coefficient for symmetric foils

In the nonlinear model, the actuator forces are represented by:

$$\boldsymbol{\tau} = [\tau_X, \tau_Y, \tau_Z, \tau_K, \tau_M, \tau_N]^T$$

and in the linear model:

$$\boldsymbol{\tau} \approx \boldsymbol{B}(t)\boldsymbol{u}(t)$$

3.5.1 Propeller Thrust

The vehicle is equipped with two electric motors connected to two counter-rotating propellers located at the stern. These actuators are capable of producing thrust in surge and if one engine is reversed while the other operate forward (differential thrust) a moment in yaw is produced. Since the propellers are counter-rotating, the torque generated by rotating the propellers is disregarded. When moving at low speeds an effect called propeller walk is also present. Propeller walk produces a lateral force in the rotational direction, due to a scooping effect. This effect is difficult to mathematically describe and depends on other factors such as hull geometry. Because of the counter-rotating propellers, propeller walk is also disregarded. As mentioned, propeller design data were used to model the engine thrust. In Table 3, the thrust for different revolutions per minute (RPM) are presented. The thrust from the starboard engine is denoted T_1, the thrust from the port engine is denoted T_2 and RPM, R_i, for the two electrical motors follow the same indexation.

Table 3: Propeller design data for one propeller

T $[N]$	R $[RPM]$
3	233
11	465
24	698
133	1628

Approximating the data in Table 3 as linear and quadratic functions gives:

$$T = 0.07017R \tag{17}$$

and

$$T = 5.1 \cdot 10^{-5} R|R| + 0.0026|R| \tag{18}$$

and can be seen in Figure 6. With Equation 18 the resulting forces generated by the propellers can be described by:

$$\tau_X = T_1 + T_2$$
$$\tau_N = -T_1 l_{py} + T_2 l_{py}$$

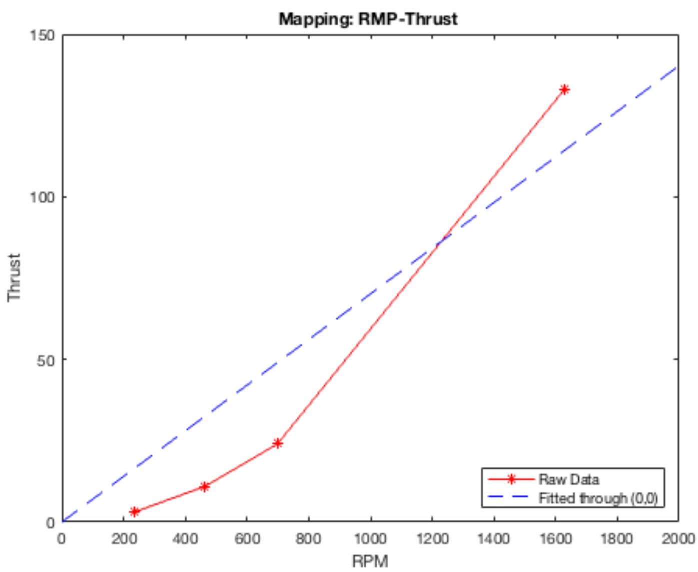

Figure 6: Thrust as a function of RPM

3.5.2 Rudders

There are two coupled rudders located on top of the hull. Because of the coupling they are approximated as one single rudder with an area twice the are of a single rudder. During forward motion u the rudder produce lift and drag. Because of the rudder position these two forces also induce moments in pitch, roll and yaw and the rudder deflection angle is denoted δ_r. The forces generated by the rudders are:

$$\begin{aligned}\tau_X &= -D_r \\ \tau_Y &= L_r \\ \tau_K &= L_r l_{roll} \\ \tau_M &= D_r l_{roll} \\ \tau_N &= L_r l_{yaw}\end{aligned}$$

3.5.3 Elevator

The elevator located at the stern of the vehicle is a control surface that produce lift and drag during forward speed u. Because of the elevator position, the lift component also induces a pitching moment and the elevator deflection angle is denoted δ_e. The forces generated by the elevator are:

$$\begin{aligned}\tau_X &= -D_e \\ \tau_Z &= L_e \\ \tau_M &= L_e l_{pitch}\end{aligned}$$

3.5.4 Elevons

The elevons are two stern planes located on each side of the vehicle. These two stern planes can be individually controlled and produce lift and drag under forward speed u. Because of the elevons position they can also induce both roll and pitch depending on the deflecting angle δ_{ev}. Lift from the starboard elevon is denoted L_{ev1} and Lift from the port elevon is denoted L_{ev2}. The forces generated by the elevons are:

$$\begin{aligned}\tau_X &= -D_{ev1} - D_{ev2} \\ \tau_Z &= L_{ev1} + L_{ev2} \\ \tau_K &= L_{ev1} l_{eroll} - L_{ev2} l_{eroll} \\ \tau_M &= L_{ev1} l_{epitch} + L_{ev2} l_{epitch}\end{aligned}$$

3.5.5 Combined Actuator Expression

Adding the forces produced by all of the actuators results in the following expression:

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_X \\ \tau_Y \\ \tau_Z \\ \tau_K \\ \tau_M \\ \tau_N \end{bmatrix} = \begin{bmatrix} T_1 + T_2 - D_r - D_e - D_{ev1} - D_{ev2} \\ L_r \\ L_e + L_{ev1} + L_{ev2} \\ L_r l_{roll} + L_{ev1} l_{eroll} - L_{ev2} l_{eroll} \\ D_r l_{roll} + L_e l_{pitch} + L_{ev1} l_{epitch} + L_{ev2} l_{epitch} \\ -T_1 l_{py} + T_2 l_{py} + L_r l_{yaw} \end{bmatrix} \tag{19}$$

As mentioned in Section 1.5, drag effects from rudders as well as some of the moments were neglected. Therefore the resulting expression for $\boldsymbol{\tau}$ used in the final model without the elevons become:

$$\boldsymbol{\tau} = \begin{bmatrix} T_1 + T_2 \\ L_r \\ L_e \\ L_r l_{roll} \\ L_e l_{pitch} \\ -T_1 l_{py} + T_2 l_{py} + L_r l_{yaw} \end{bmatrix} \tag{20}$$

3.6 Equations of Motion

By combining Equation 9, 10, 11, 13 and 20 all terms in the Equation 6 are described and can be solved for the acceleration $\dot{\boldsymbol{\nu}}$. Reorganizing the terms and left multiplying with $\boldsymbol{M^{-1}}$ gives the following expression:

$$\dot{\boldsymbol{\nu}} = \boldsymbol{M}^{-1}(-\boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} - \boldsymbol{g}(\boldsymbol{\eta}) - \boldsymbol{g_0} + \boldsymbol{\tau}) \tag{21}$$

3.7 Parameter Estimation

Since both the dynamics and the stability of the created models depends on a large number of hydrodynamic derivatives it is important to derive or estimate sufficiently accurate values for each parameter. In this thesis the main focus was to estimate sufficient values to the linear damping coefficients necessary for a linear model representation. Using the VP model described in Section 3.9.1 each DOF was excited individually, either by external forces or with the vehicle's own actuators and the time response was compared to vehicle behavior during live testing and the data acquired from these tests. The damping in surge was derived with the assumption that the vehicle had a maximum speed of $2.5\,m/s$. The results from these estimations are presented in Section 5.

3.8 nonlinear State-Space

Combining Equation 2 and 21 results in a nonlinear state-space on the form:

$$\boxed{\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) = \boldsymbol{f}(t, \boldsymbol{\nu}, \boldsymbol{\tau})}$$

where $\dot{\boldsymbol{x}} = [\dot{\boldsymbol{\eta}}, \dot{\boldsymbol{\nu}}]^T$. This model works well for nonlinear control applications, but since it is expressed in both NED and BODY coordinates it is not suitable for linearization. Instead, other methods described in Section 3.9, 3.9.1 and 3.9.2 were used.

3.9 Linearization

Using the standard technique described in Equation 22 the equations of motion can be transformed into a nonlinear state-space expressed in only BODY coordinates, which is better suited for linearization. The resulting system is of order 12 according to the DOF definition presented in Section 2.2 and can from here be linearized using different techniques.

$$\boldsymbol{x} = \begin{bmatrix} \int_0^\infty \boldsymbol{\nu} dt \\ \boldsymbol{\nu} \end{bmatrix} \iff \dot{\boldsymbol{x}} = \begin{bmatrix} \boldsymbol{\nu} \\ \dot{\boldsymbol{\nu}} \end{bmatrix} \tag{22}$$

3.9.1 Vessel parallel coordinates LTI-Model

One technique used to derive the linearized equations of motions during this thesis is Vessel parallel coordinates (VP). VP coordinates use the assumption that ϕ and θ are small which is a good approximation for a metacentric stable [6] AUV.

By rotating the body axis at each time step one gets:

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J_\Theta}(\boldsymbol{\eta})\boldsymbol{\nu} \overset{\phi\,=\,\theta\,=\,0}{\approx} \boldsymbol{P}(\boldsymbol{\Psi})\boldsymbol{\nu}$$

where

$$\boldsymbol{P}(\boldsymbol{\Psi}) = \begin{bmatrix} \boldsymbol{R}(\boldsymbol{\Psi}) & \boldsymbol{0_{3x3}} \\ \boldsymbol{0_{3x3}} & \boldsymbol{I_{3x3}} \end{bmatrix}$$

and

$$\boldsymbol{R}(\boldsymbol{\Psi}) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is the rotation matrix in yaw and a VP coordinate system can be defined as:

$$\boldsymbol{\eta_p} := \boldsymbol{P^T}(\boldsymbol{\Psi})\boldsymbol{\eta} \tag{23}$$

Assuming that the vehicle will operate at a low speed of maximum 2 m/s gives the yaw rate $r \approx 0$ and Equation 23 can be further reduced to:

$$\dot{\boldsymbol{\eta}}_{\boldsymbol{p}} \approx \boldsymbol{\nu}$$

which is a linear equation in ν. Further assuming that the vehicle is neuatrally boyant ($W = B$) and that the center of gravity (CG) and the center of buoyancy (CB) has the same x and y positions ($x_g = x_b$) and ($y_g = y_b$), Equation 13 representing the restoring forces is reduced to:

$$\boldsymbol{G} = diag\{0, 0, 0, (z_g - z_b)W, (z_g - z_b)W, 0\}$$

For low speed applications $\boldsymbol{C(0)} \approx \boldsymbol{0}$, $\boldsymbol{D_n(0)} \approx \boldsymbol{0}, \boldsymbol{\tau} \approx \boldsymbol{Bu}$ and since the VBS has not been implemented yet $\boldsymbol{g_0} = \boldsymbol{0}$. This results in a linearized state-space model on the form:

$$\dot{\boldsymbol{\eta}}_{\boldsymbol{p}} = \boldsymbol{\nu}$$
$$\dot{\boldsymbol{\nu}} = \boldsymbol{M^{-1}}(-\boldsymbol{D\nu} - \boldsymbol{G\eta_p} + \boldsymbol{Bu})$$

This is a linear time-invariant (LTI) model on the form $\dot{\boldsymbol{x}}(t) = \boldsymbol{Ax}\ (t) + \boldsymbol{Bu}(t)$ where:

$$\boldsymbol{u}(t) = \begin{bmatrix} \boldsymbol{u_1} \\ \boldsymbol{u_2} \\ \boldsymbol{u_3} \\ \boldsymbol{u_4} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R_1} \\ \boldsymbol{R_2} \\ \boldsymbol{\delta_r} \\ \boldsymbol{\delta_e} \end{bmatrix}$$

$\boldsymbol{R_1}$ is the RPM of engine one and $\boldsymbol{R_2}$ is the RPM of engine two.

3.9.2 Jacobian LTV-Model

Another method for linearization that is better suited for deriving linear time-varying (LTV) models is to use a Jacobian matrix [10]. The resulting state-space decription has the form:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{J_A}(t)\boldsymbol{x}(t) + \boldsymbol{J_B}(t)\boldsymbol{u}(t)$$

where

$$\boldsymbol{J_A}(t) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_{12}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_{12}}{\partial x_1} & \cdots & \frac{\partial f_{12}}{\partial x_{12}} \end{bmatrix} \Bigg|_{\boldsymbol{(x,u)=(x(t),u(t))}}$$

and

$$\boldsymbol{J_B}(t) = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_4} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_{12}}{\partial u_1} & \cdots & \frac{\partial f_{12}}{\partial u_4} \end{bmatrix} \Bigg|_{\boldsymbol{(x,u)=(x(t),u(t))}}$$

This model is similar to the LTI-model derived by a VP coordinate frame in Section 3.9.1, but with the difference that this model can be re-evaluated for current values of $\boldsymbol{x}$ and $\boldsymbol{u}$ thus creating a LTV model. Using a Jacobian on the nonlinear equations of motion also allows for different model complexities, where certain applications require more or less accurate models.

3.10 Discretization

To make a MPC implementation possible, the last step in the modeling process is to transform the continuous time model into a discrete time model on the form $\boldsymbol{x}[t+1] = \boldsymbol{A}[t]\boldsymbol{x}[t] + \boldsymbol{B}[t]\boldsymbol{u}[t]$. Like in the continuous time case, this model can be both time variant and invariant, and both types were used during this thesis. There exists several different techniques for discretization of a continuous time system. Due to time constraints and since the discretization was not considered as important as the MPC implementation itself, Matlab's built in continuous to discrete (C2D) function together with a sampling time h of 0.2 s was used.

4 Control

The third phase of this thesis covers control aspects. The ultimate goal was to achieve MPC reference tracking using a LTV model describing the complete dynamics of the vehicle. To reach this goal there are a number of necessary steps that need to be taken. The stability, controllability and observability needs to be evaluated to ensure that solutions can be derived and that the control problem is feasible. In this phase the elevons are not considered a part of the physical system and therefore not included as actuators in the controllers, resulting in a total of 4 actuators.

4.1 Existing System

The vehicle's existing control system uses standard PID controllers, either working as individual units or connected into cascade controllers. Even though this solution is simple and performs well, there are some major drawbacks. Each output needs its own controller which makes the software structure more complicated than necessary. Each controller also works under the assumption of decoupled motions, which can lead to performance limitations or contradicting control objectives. Tuning all these individual controllers can also be a problem. Every time the vehicle configuration is changed all of the controllers have to be retuned.

4.2 Model Analysis

Before any type of control implementation is created model analysis is usually performed. This analysis often consist of three steps. Stability, controllability and observability. The purpose of these tests are to find out what is physically possible to do in terms of control, regulation and estimation. If the results from these tests are not satisfactory, there usually is no point in continuing with the control implementation. Instead, more work with modeling or other techniques such as the use of observers need to be employed to ensure that the desired control objectives can be met.

4.2.1 Stability

Stability is one of the fundamental principles in control engineering [11]. Even though controllers can be designed that stabilize unstable systems, the stability of a system is important because it defines how the system behaves in terms of solutions, equilibria and whether a bounded input generates a bounded output (BIBO). The poles and zeros of a system can further be analyzed to determine other physical limitations such as bandwidth, frequency response and the highest achievable controller performance.

There are several methods that can be used to determine whether a system is stable or not but during this thesis standard methods for evaluating eigenvalues of the system matrix $\boldsymbol{A}$ were used. The system equilibria was also considered by looking at $\mathbf{0} = \boldsymbol{A}(t)\boldsymbol{x}(t) + \boldsymbol{B}(t)\boldsymbol{u}(t)$. This equation shows that local equilibrium exists everywhere in $\mathbb{R}^3$ thus making this a pure regulation problem.

During live tests of the vehicle's basic systems and functions during Autumn 2019, both stability and system equilibria were inferred from dynamic and static behavior in the water. Even though this is a visual confirmation it is still a strong stability proof.

4.2.2 Controllability

The concept of controllability denotes whether a controller can drive the states of the system to any arbitrary point in $\mathbb{R}^3$ or not, also refereed to as the controllable subspace [13]. Since the goal of this thesis is reference tracking to arbitrary way points controllability is of great importance and to ensure that feasible solutions exist to any waypoint, the controllable subspace need to be equal to $\mathbb{R}^3$. To check for controllability, the standard method where $rank(\mathcal{C})$ must be equal to the system order was used:

$$\mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$

Results from this test will be presented in Section 5.

4.2.3 Observability

The concept of observability denotes whether all of the internal system states can be inferred from external outputs or not [12]. Observability was evaluated using the the standard test for observability were $rank(\mathcal{O})$ must be equal to the system order:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

With this test it can be shown that $rank(\mathcal{O})$ < the order of the system thus making it unobservable. However, the vehicle itself is equipped with sensors capable of measuring all the 12 states in real-time which completely removes this problem since state values can always be measured. If these sensor measurements would not have been available, other methods including observers [13] would have been needed.

Even though observability does not impose a problem for the control implementation it introduces another concept previously not mentioned. The output of a system, described by $\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x}(t)$, shows which quantities are of importance and is often used in regulation applications such as integral states described in Section 4.3. In general a system of square form is desired [14] which leads to $\boldsymbol{C} \in \mathbb{R}^{4\text{x}12}$ to ensure that the number of inputs are equal to the number of outputs. Since the objective is to regulate position, x, y and z are the obvious choices leaving one output to be determined. More about this last output will be presented in Section 6. The output matrix has the following form:

$$\boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ . & . & . & . & . & . & . & . & . & . & . & . \end{bmatrix}$$

where the row of dots represent the last undetermined output and any linear state combination can be chosen.

4.3 Model Augmentations

In order to regulate a system to an equilibrium state other than its global equilibrium the dynamics model needs to be augmented. There are two commonly used methods to achieve offset free tracking [15].

The first method is called error tracking and requires a model augmentation where the state $\boldsymbol{x}$ is remapped into $\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}_{target}$. In practice this results in the system equilibrium being shifted to an augmented equilibrium $\boldsymbol{x}_{target}$ and a feedback regulator will drive the system to this value. For a linear system on the form $\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t)$ this remapping is simple and takes the form [15]:

$$\dot{\boldsymbol{e}}(t) = \boldsymbol{A}\boldsymbol{e}(t) + \boldsymbol{B}\boldsymbol{u}(t)$$

where $\boldsymbol{u}(t) = \boldsymbol{-K(x - x_{target})}$ and $\boldsymbol{K}$ is the feedback gain used to shape closed loop behavior.

The second method is called integral action and requires a model augmentation where an integral state $\boldsymbol{\dot{x}_i = y - r = Cx - r}$ representing the output error is introduced, where $\boldsymbol{r = y_{target}}$. In practice this tells the controller that the only equilibrium is when the output is equal to the reference, thus creating integral action and the controller can produce the necessary input required to reduce the output error to 0. This type of augmentation also provide robustness against model uncertainties and disturbances. The augmented state-space takes the following form [15]:

$$\begin{bmatrix} \dot{\boldsymbol{x}} \\ \dot{\boldsymbol{x}}_{\boldsymbol{i}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{0} \\ \boldsymbol{C} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{x}_{\boldsymbol{i}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{0} \end{bmatrix} \boldsymbol{u} - \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{r}$$

$$\boldsymbol{y} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{x}_{\boldsymbol{i}} \end{bmatrix}$$

4.4 Linear Control Methods

Linear control methods include state-feedback control solutions to systems on the form $\dot{\boldsymbol{x}}(t) = \boldsymbol{A}(t)\boldsymbol{x}(t)+\boldsymbol{B}(t)\boldsymbol{u}(t)$, where the solution has the form $\boldsymbol{u}(t) = -\boldsymbol{Lx}$ (t).

4.4.1 State-Feedback

State-feedback, or pole placement as it is often referred to, is usually the first step when employing linear control methods. This method exploits the property that the eigenvalues of the system matrix $\boldsymbol{A}(t)$ corresponds to the open loop poles of the system. Feeding back $\boldsymbol{u}(t) = -\boldsymbol{Kx}$ (t) allows a designer to place the closed loop poles of $\dot{\boldsymbol{x}}(t) = (\boldsymbol{A}(t) - \boldsymbol{B}(t)\boldsymbol{K})\boldsymbol{x}(t)$ in any arbitrary position provided that the system is controllable [13]. Since the pole placement for stable systems is known this method is suitable for both regulation and stabilization and creates understanding of what state-feedback control is trying to achieve. However, this method is rarely used in practice since state-feedback only addresses stability and not performance. Performance analysis requires investigation of e.g. time or frequency responses for each individual pole. Instead some of the more advanced methods described in Section 4.4.2 and 4.5 can be used.

4.4.2 LQR

A linear-quadratic regulator (LQR) is an optimal state-feedback controller that uses linear constraints and a quadratic cost function to operate a target system at a minimal cost [16]. The advantage of working with LQR control instead of state-feedback is that when the control horizon N is allowed to reach infinity (infinite horizon LQR) [12] the optimization converges and the solution from the algebraic Riccati equation (ARE) can be used to derive the optimal state-feedback. Another advantage with this method is that effort is usually shifted from optimizing the controller performance to weighting different performance objectives. The proof for this method will be left out, but for a continuous-time linear-system and a quadratic cost function $\boldsymbol{J}$ on the form:

$$\begin{aligned} \min_{u} \quad & \boldsymbol{J} = \int_0^\infty \boldsymbol{x^T Q_1 x + u^T Q_2 u} \\ \text{s.t} \quad & \dot{\boldsymbol{x}} = \boldsymbol{Ax + Bu} \end{aligned}$$

Where $\boldsymbol{Q_1}$ and $\boldsymbol{Q_2}$ are matrices defining the state and input weights. The optimal solution is given by:

$$\boldsymbol{u = -Kx}$$

where

$$\boldsymbol{K = Q_2^{-1} B^T P}$$

is the optimal gain and $\boldsymbol{P}$ is chosen as the solution to the ARE:

$$\boldsymbol{A^T P} + \boldsymbol{PA} - \boldsymbol{PBR^{-1}B^T P} + \boldsymbol{Q} = \boldsymbol{0}$$

Solving LQR problems numerically is challenging but efficient solvers that can handle these structures exists and the hardest task is to weigh the cost function and augment the model to behave according to the control objective.

Since this formulation can not handle additional state or input constraints there exists one major drawback. The optimal gain $\boldsymbol{K}$ does not consider physical limitations on inputs and therefore the input $\boldsymbol{u} = -\boldsymbol{Kx}$ has to be manually saturated.

4.5 MPC

Model predictive control, or receding horizon LQR with constraints [12], is an advanced control method where a cost function $\boldsymbol{J}$ is minimized over a fixed horizon $\boldsymbol{N}$ subject to the discrete time system dynamics $\boldsymbol{x}[t+1] = \boldsymbol{Ax}\,[t] + \boldsymbol{Bu}\,[t]$. When the optimal input sequence $\boldsymbol{u^*} \in \mathrm{I\!R}^{\mathbf{N-1}}$ is obtained, the first control action $\boldsymbol{u_1^*}$ is taken and the entire optimization is repeated. The basic formulation is similar to the LQR controller presented in Section 4.4.2, but the use of discrete dynamics and an iterative finite horizon allows additional state and input constraints to be implemented directly in the optimization.

4.5.1 Basic Formulation

There are several ways to formulate a MPC algorithm depending on whether the cost function is affine in $\boldsymbol{u}$ and how the terminal set $\boldsymbol{x_N}$ is handled to ensure future stability and feasibility [12]. In this thesis, the formulation presented in Equation 24 together with different model augmentations presented in Section 4.5.2 was used.

$$\begin{aligned}
\min_{u} \quad & \sum_{\boldsymbol{t=0}}^{\boldsymbol{N-1}} \boldsymbol{x_t^T Q_1 x_t} + \boldsymbol{u_t^T Q_2 u_t} + \boldsymbol{x_N^T Q_N x_N} \\
\text{s.t} \quad & \boldsymbol{x}[t+1] = \boldsymbol{Ax}\,[t] + \boldsymbol{Bu}\,[t] \\
& |u_{1,\boldsymbol{t}}| \leq 500 \\
& |u_{2,\boldsymbol{t}}| \leq 500 \\
& |u_{3,\boldsymbol{t}}| \leq 0.3491 \\
& |u_{4,\boldsymbol{t}}| \leq 0.3491
\end{aligned} \tag{24}$$

4.5.2 Additional Augmentations

When working with MPC regulation and tracking there are some problems that need to be addressed. Forcing the output $\boldsymbol{y}$ to a reference point $\boldsymbol{r}$ leads to the output error relation:

$$\boldsymbol{e_t = r_t - Cx_t \rightarrow 0}$$

steady state $\boldsymbol{e_k = 0}$ is desired but to achieve this, $\boldsymbol{u_k \neq 0}$ which is not a steady state. One way to deal with this is to augment the system using input increments [17].

Let:

$$\boldsymbol{u_t = u_{t-1} + \Delta u} \tag{25}$$

and define $\boldsymbol{u_{t-1}}$ as a new state variable and $\boldsymbol{\Delta u}$ as the new control input. This leads to an augmented formulation with integral action on the input that circumvents the steady-state problem on the form:

$$\begin{aligned}
\min_{\Delta u} \quad & \sum_{t=0}^{N-1} \boldsymbol{(r - Cx)_t^T Q_1 (r - Cx)_t + \Delta u_t^T Q_2 \Delta u_t} \\
\text{s.t} \quad & \begin{bmatrix} \boldsymbol{x_{t+1}} \\ \boldsymbol{u_{t+1}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{x_t} \\ \boldsymbol{u_t} \end{bmatrix} + \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{\Delta u_t}
\end{aligned} \tag{26}$$

This is now a state-space model of order 16, where:

$$\boldsymbol{y_t} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{x_t} \\ \boldsymbol{u_t} \end{bmatrix} \tag{27}$$

To ensure offset-free tracking integral states may need to be introduced on the state variable $\boldsymbol{x}$ as well. Using a method similar to the one described in Equation 25 and 26 the system can be described in both state and input increments as described in [18] and [19].

Let:

$$\boldsymbol{\Delta x_{t+1} = A \Delta x_t + B \Delta u_t} \tag{28}$$

where $\boldsymbol{\Delta x_t = x_t - x_{t-1}}$. The output $\boldsymbol{y_t}$ need to be connected to $\boldsymbol{\Delta x_k}$ and $\boldsymbol{y_t = Cx_t}$ gives:

$$\boldsymbol{y_t = y_{t-1} + C \Delta x_t} \tag{29}$$

Combining Equation 28 and 29 and choosing:

$$\boldsymbol{\bar{x}} = \begin{bmatrix} \boldsymbol{\Delta x_t} \\ \boldsymbol{y_{t-1}} \end{bmatrix}$$

as the new state vector gives the state space model:

$$\begin{bmatrix} \boldsymbol{\Delta x_{t+1}} \\ \boldsymbol{y_t} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{0} \\ \boldsymbol{C} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Delta x_t} \\ \boldsymbol{y_{t-1}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{0} \end{bmatrix} \boldsymbol{\Delta u_t} \tag{30}$$

This is also a state-space model of order 16 that can replace the linear constraints in Equation 26 to create integral action on both state and input, where:

$$\boldsymbol{y_t} = \begin{bmatrix}\boldsymbol{C} & \boldsymbol{I}\end{bmatrix}\begin{bmatrix}\boldsymbol{\Delta x_k}\\ \boldsymbol{y_{t-1}}\end{bmatrix} \tag{31}$$

The augmented MPC formulation described by Equation 30 has one weakness. Augmenting both states and inputs with increments, without adding states that describe how $\boldsymbol{u}$ is affected by $\boldsymbol{\Delta u}$ removes the option to constrain the input as presented in Equation 24. This can be handled in two ways. Either 4 additional states that describe the relation between $\boldsymbol{u}$ and $\boldsymbol{\Delta u}$ are added, which directly enables constraints on $\boldsymbol{u}$ with the downside of increasing the state-space order to 20. The other option is to let the Matlab algorithm keep track of the true value of $\boldsymbol{u}$ independent from the state-space model and use this information to convert limitations on $\boldsymbol{u}$ to limitations on $\boldsymbol{\Delta u}$ and constrain these in the MPC formulation.

5 Results

The number of steps taken from unmodelled vehicle to controller evaluation has produced several different results. From all the theory and work with modeling, a generator that can produce symbolic models has been created. This model generator takes simplifications and assumptions such as axes of symmetry and whether the vehicle is neutrally buoyant or not as input and produces a full nonlinear model according to these conditions.

With the model generator and the linearization techniques described in Section 3, two linear state-space models on the form $\dot{\boldsymbol{x}}(t) = \boldsymbol{A}(t)\boldsymbol{x}(t) + \boldsymbol{B}(t)\boldsymbol{u}(t)$ representing the nonlinear dynamics of LoLo were created. The system matrix for the model derived with the use of a VP reference frame can be seen in Equation 32.

$$\boldsymbol{A} = \begin{bmatrix} 0&0&0&0&0&0&1&0&0&0&0&0 \\ 0&0&0&0&0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&0&0&0&0&1&0&0 \\ 0&0&0&0&0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&-0.0667&0&0&0&0&0 \\ 0&0&0&0&0&0&0&-8.33&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-10&0&0&0 \\ 0&0&0&-32.2&0&0&0&0&0&-10.94&0&0 \\ 0&0&0&0&-3.18&0&0&0&0&0&-1.62&0 \\ 0&0&0&0&0&0&0&0&0&0&0&-3.86 \end{bmatrix} \tag{32}$$

The model generated with the use of a Jacobi matrix has a sparse structure similar to the one presented in Equation 32. The difference with this Jacobi matrix model is that additional couplings between DOFs may be introduced in the nonlinear representation which allows that a linearization of varying complexity can be produced.

In Section 3.5.1 two functions for the propeller thrust are presented and they can be seen in Figure 6. In the electrical motors operating range of -500 to 500 RPM, Equation 17 representing linear thrust fitted through 0 has a significantly higher value than the nonlinear thrust represented in Equation 18. Therefore, Equation 18 linearized around 500 RPM are used in the final dynamics model.

The results from the standard test for controllability indicates that the vehicle is controllable under forward speed ($u > 0$), but not while standing still ($u = 0$). Since rudders and foils only produce forces during forward motion this was expected, but this resulted in some changes to the structure of the input matrix $\boldsymbol{B}(t)$. The resulting input matrix can be seen in Equation 33 where actuator forces are linearized around $(R, u) = (500, 1)$. The result of this linearization is that actuators are assumed to produce the same amount of force independent of the vehicle's velocity in surge u.

$$\boldsymbol{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.0002 & 0.0002 & 0 & 0 \\ 0 & 0 & 0.0212 & 0 \\ 0 & 0 & 0 & 0.0191 \\ 0 & 0 & 0.0365 & 0 \\ 0 & 0 & 0 & 0.0260 \\ -3.6\times10^{-5} & 3.6\times10^{-5} & 0.0202 & 0 \end{bmatrix} \tag{33}$$

For these models to be useful in controller design and simulation the hydrodynamic derivatives describing linear damping had to be determined. The estimated parameter values can be found in Table 4 and were derived according to the description in Section 2.6. All simulations of dynamic behavior during this thesis was performed using the Euler forward method:

$$\boldsymbol{x}(t+h) = \boldsymbol{x}(t) + \dot{\boldsymbol{x}}(t)h$$

where h is a time step and $(t+h) \approx (t+1)$ when working with discrete systems and dynamics.

Since the hydrodynamic derivatives influence both stability and dynamic behavior their values are of great importance. In Figure 7 - 12 the time response for surge, sway, heave, pitch, roll and yaw rate using the parameter values in Table 4 are presented. From the time response, stability and boundedness can also be inferred for each DOF since the vehicle returns to its equilibrium after subjected to external disturbances and the parameter values creates an upper bound on the velocities.

Table 4: Estimated linear damping coefficients

X_u	=	-20
Y_v	=	-2500
Z_w	=	-3000
K_p	=	-400
M_q	=	-600
N_r	=	-1500

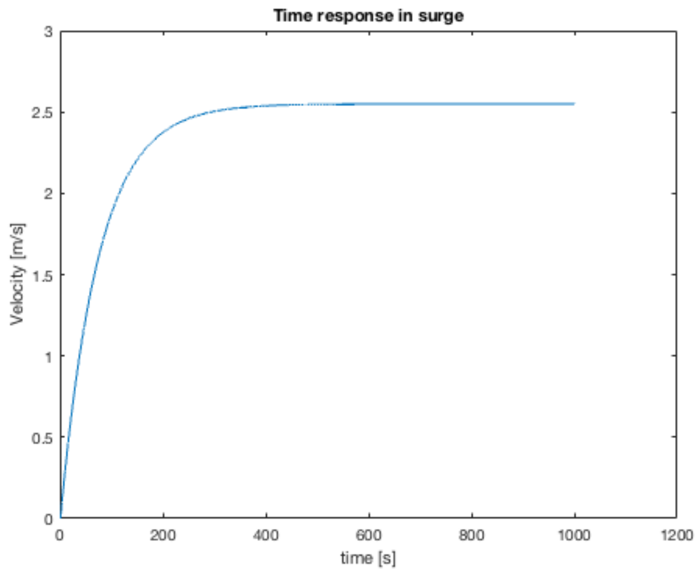

Figure 7: Time response in surge when both engines operate at 500 RPM

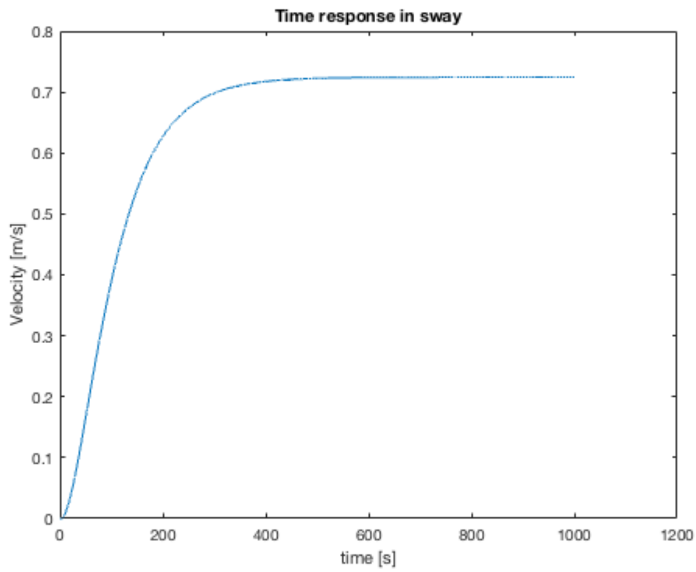

Figure 8: Time response in sway when both engines operate at 500 RPM and the rudder deflection is 0.1745 radians

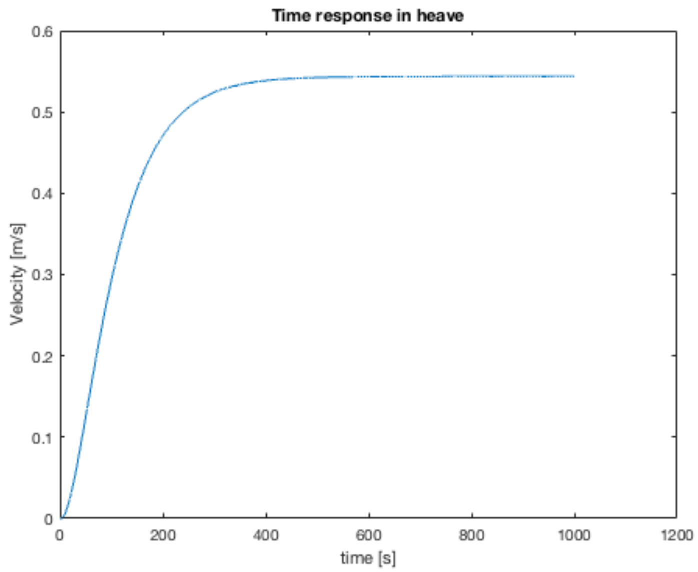

Figure 9: Time response in heave when both engines operate at 500 RPM and the elevator deflection is 0.1745 radians

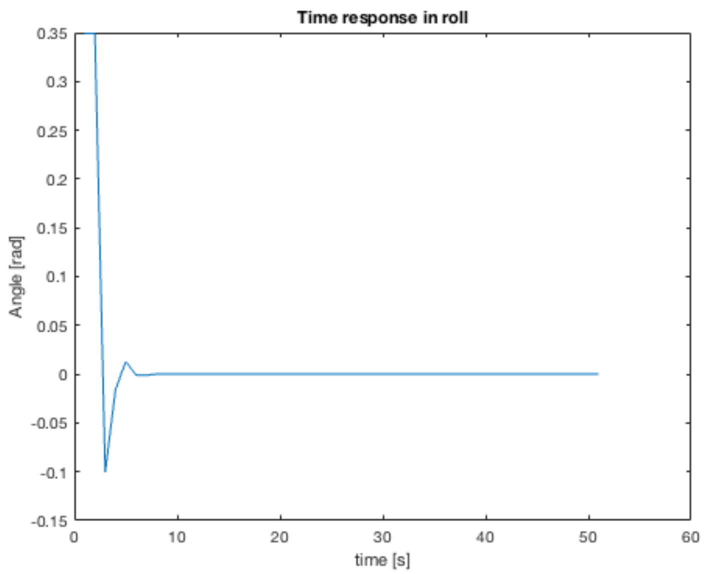

Figure 10: Time response in roll when forced 0.3491 radians from equilibrium

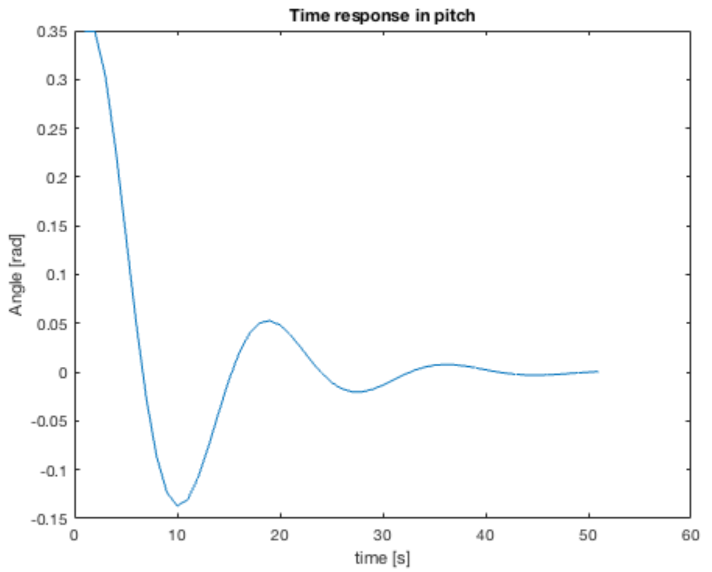

Figure 11: Time response in pitch when forced 0.3491 radians from equilibrium

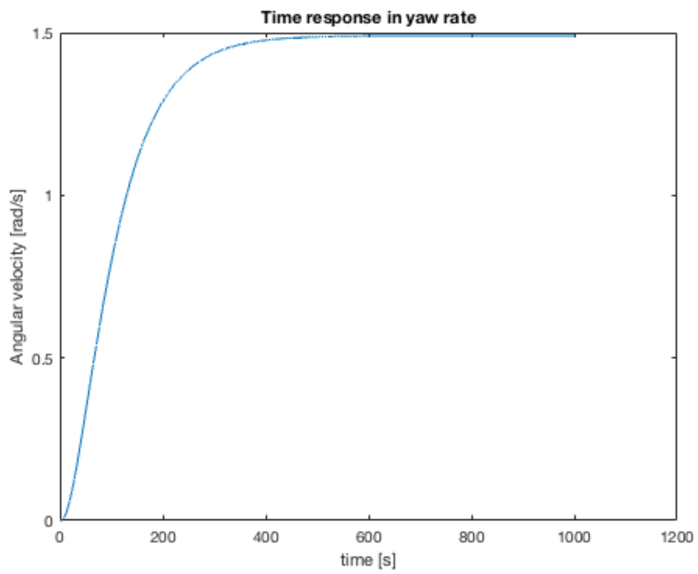

Figure 12: Time response in yaw when both engines operate at 500 RPM and the rudder deflection is 0.1745 radians

During LQR simulation the effect of different state and input weights were investigated. In Equation 34 and 35 the weight matrices used in the final LQR controller are presented.

$$\boldsymbol{Q_1} = diag\{100, 10, 10, 1, 1, 0.1, 1, 1, 1, 1, 1, 1\} \tag{34}$$
$$\boldsymbol{Q_2} = diag\{10, 10, 1, 1\} \tag{35}$$

When the closed loop system was simulated with the goal of off-set free tracking the final undetermined output mentioned in Section 4.2.3 was chosen to output yaw.
In Figure 13 the simulated vehicle trajectory in North-East coordinates using LQR tracking and the linear VP model can be seen. This controller is derived according to the theory in Section 4.4.2 and uses an error-tracking augmentation as described in Section 4.3. The vehicle starts in $(0, 0)$ marked with a yellow dot and the target waypoint $\boldsymbol{x_{target}}$ is $(100, 100)$ marked by a black dot.

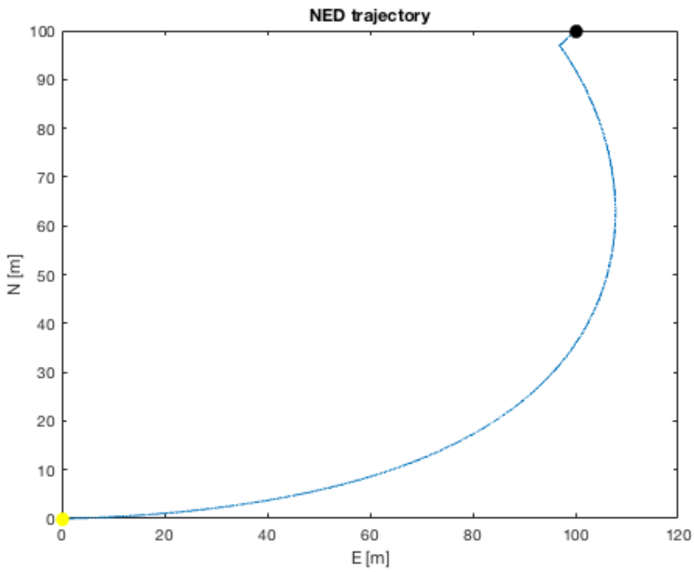

Figure 13: Simulated LQR trajectory

In Figure 14 the simulated vehicle trajectory in north-east coordinates using MPC tracking and the linear VP model can be seen. This controller is derived according to Section 4.5 and use integral action on both states and inputs as described in Section 4.5.2. The vehicle starts in $(0, 0)$ marked with a yellow dot and the target waypoint $\boldsymbol{x_{target}}$ is $(100, 100)$ marked by a black dot.

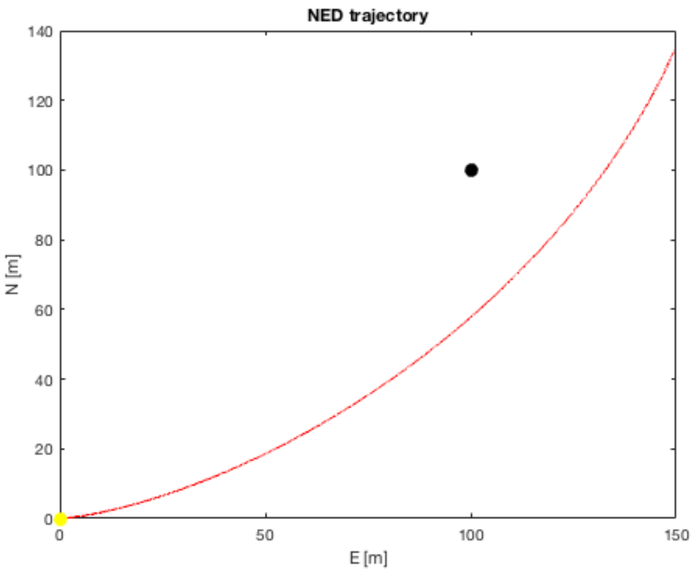

Figure 14: Simulated MPC trajectory

During the simulation in Figure 14 the vehicle travels a distance of 201 m during a time of 400 s. This results in an average surge velocity of 0.5 $^{m}/_{s}$ and the computation time for the MPC controller during 2000 repeated horizons was approximately 23 s which correspond to an average calculation time per horizon of 0.0115 s.

6 Analysis and Discussion

When the nonlinear dynamics model was derived it was decided that creating a full-scale model including all the couplings and dynamical effects and subsequent down-scaling was more beneficial than starting from a simple model with subsequent up-scaling. The reason for this is that the full-scale model is a suitable representation of the physical system whereas the simple model is an approximation. This is important, since the linear models derived later on become either a linearization of an accurate model or a linearization of an approximate model. In practice this makes a big difference for how model dependent results can be interpreted. Another benefit with the creation of a full-scale model is that this model can be used for other purposes such as simulation.

During the modeling phase the ambition was first to place the coordinate frame origin in the bow of the vehicle. The reason for this was that during motion there is not much use in knowing the distance from an obstacle to a point located inside the vehicle. Placing the coordinate frame in the bow provides the true distance to an obstacle. The downside with this placement is that the expression for the equations of motion become increasingly complex and dynamical behavior become harder to evaluate. Because of this, the coordinate frame origin was changed from the bow of the vehicle to coincide with CG. The benefits with this location is that rotational and translational components due to a shift of coordinate frame origin is avoided. Additionally, a coordinate frame located in CG is what allowed for other simplifications e.g. the ones used in Section 3.4, where Equation 12 could be reduced to Equation 13.

As mentioned in Section 1, this thesis focus on control methods using linear models. There are several reasons for this, the main one being the time required to derive the control inputs. Using a linear model to describe the dynamic behavior of a nonlinear system will reduce the model accuracy but the upside is that the calculation time for advanced methods such as MPC is drastically reduced. This creates a trade-off between model accuracy and the calculation time needed by the controller. Both accuracy and calculation time is important for the implementation in a physical system due to limited computational power. If the calculation time is to long the controller will not have enough time to derive the necessary input required at time t and a delay is introduced into the system. This delay limits performance and can cause instability. Model accuracy is important for robustness and to make sure that the derived solutions work in the physical system.

The modeling technique described by [6] where the state-space model is described in both NED and BODY coordinates work for nonlinear control applications, but is not suitable for linear applications. Since $[N, E, D] \neq [x, y, z]$ and the mapping between these coordinates is nonlinear a sufficient linear model with mixed coordinates is hard to derive. As described in Section 3.9 a linear model expressed in only BODY coordinates has been used which have caused a number of difficulties. Tracking a waypoint in the global NED frame can be seen as a constant reference. However, in the BODY frame this reference is time-varying and depends on x_b, y_b, z_b and ψ. This adds a level of complexity in the controller when the reference has to be remapped into a new coordinate frame. In the LQR solution no impact from this mapping have been discovered in the results but implementing this reference map in the MPC algorithm proved difficult and might be one reason to the low tracking performance generated by MPC.

There are two main reasons why a MPC solution is more beneficial than a LQR solution. The MPC framework have the ability to handle constrained optimization and because of this the controller can directly handle physical limitations on states and inputs unlike LQR where signals have to be physically limited before implemented. Since MPC is a receding horizon online algorithm a type of feedback mechanism is also implemented and some capability to handle prediction errors due to discrepancies between the system model and the physical system is introduced. As mentioned in Section 1 calculation time for the MPC controller was of concern. Receding horizon constrained optimization is far more demanding than infinite horizon LQR and for the MPC controller to be feasible in practise the calculation time for each horizon needs to be shorter than the discrete sampling time h. As presented in Section 5 the calculation time per horizon was 0.0115 s when a sampling time of 0.2 s was used making this controller feasible in terms of calculation time with the used hardware.

The effect of the discretization technique used during this thesis was not studied. Discretization technique and sampling time are both factors that can effect the result, dynamic behavior and needs to be investigated to ensure that the discrete time linear system is a good representation of continuous time dynamics. The order in which linearization and discretization were performed might also affect the result but since the goal was to investigate control models and not validate dynamics models this result is left as future work.

The adaption of control surfaces such as rudders, elevator and elevons for actuation instead of electrical motors that can generate thrust in different directions is an active design choice. Control surfaces are more energy efficient than electrical motors which is beneficial for the energy consumption and increase mission duration. The downside is that all the produced forces are speed dependent and the existing configuration makes LoLo an underactuated AUV. While forward speed makes the vehicle controllable as described in Section 4.2.2 underactuation is a physical property and since no independent forces can be produced in either sway or heave a control objective in these DOFs is hard to satisfy. To much focus on controllability while not fully respecting underactuation might be one of the reasons to the low tracking performance generated by MPC. A way to deal with this problem is presented in Section 8.

Another factor that concern a physical control implementation is model validation. To ensure that the derived controllers produce sufficient tracking performance in a physical implementation the accuracy of the dynamics models needs to be evaluated. The hydrodynamic parameters used in this thesis are estimates and constitute a fraction of the parameters required to fully describe the non-linear dynamic behaviour of an AUV. Because of this the true performance of an implemented controller can only be verified in physical testing or with a perfectly accurate dynamics model with all hydrodynamic parameters determined. The results derived in this thesis can therefore only be used as proof of concept and to deduce feasibility in terms of performance and suitability.

Since θ and ϕ are approximated to 0 in the linear model, one dynamic effect is completely neglected in the dynamics model. This dynamic effect is a consequence of that the vehicle hull itself work as control surface when the angle of attack (AOA) is not zero. The projected hull-area is significantly larger than any of the control surfaces and because of this large forces will be produced when the AOA is different from zero. These forces might create a problem in a physical implementation and have to be further evaluated.
In Section 4.2.3 the output of the system was introduced and a free choice of the fourth and last output was mentioned. Since the vehicle is underactuated and therefore have a hard time fulfilling a control objective in 6 DOFs, ψ was chosen as the last output. The reason for this is that underactuation in sway can be compensated by actuation in yaw and choosing yaw as the final output allows the controllers to track desired angular orientation and the vehicle can reach any desired waypoint.

7 Conclusion

All the results from this thesis indicates that waypoint tracking with optimal control using a linear dynamics model is feasible but also model dependent. To ensure that a physical implementation is possible either physical tests with the derived controllers or model validation including derivation of accurate hydrodynamic parameters is required. Another conclusion that can be inferred from that LoLo is underactuated and the results of this thesis is that this vehicle might not be able to utilize the full effect of an advanced controller like MPC and simpler methods like LQR might be a more viable choice.

8 Future Work

To further improve the results and work produced by this thesis there are at least four possible routes one may take. Common for all of these four routes are that extensive work with the vehicle model is required. While the full model created in this thesis is a complete description of the AUV dynamics, work to determine all the hydrodynamic derivatives is still needed to ensure that the model provides an accurate representation of the physical system. Before the dynamics model is properly evaluated, no results from the derived controllers can be trusted to solve the tracking problem in a physical implementation.

To circumvent the problem of waypoints and vehicle dynamics being described in different reference frames, an approach described by Equation 36 through 45 can be used. The benefit with this model is that it describes the vehicle dynamics completely in the NED and the reference waypoints can be directly tracked. The dowside however, is that additional nonlinear couplings are introduced and creating a sufficient linear representation becomes more difficult.

$$\boldsymbol{M}^*(\boldsymbol{\eta})\ddot{\boldsymbol{\eta}} + \boldsymbol{C}^*(\boldsymbol{\nu},\boldsymbol{\eta})\dot{\boldsymbol{\eta}} + \boldsymbol{D}^*(\boldsymbol{\nu},\boldsymbol{\eta})\dot{\boldsymbol{\eta}} + \boldsymbol{g}^*(\boldsymbol{\eta}) + \boldsymbol{g}_0^*(\boldsymbol{\eta}) = \boldsymbol{\tau}^* \tag{36}$$

Where

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J}_{\Theta}(\boldsymbol{\eta})\boldsymbol{\nu} \tag{37}$$
$$\ddot{\boldsymbol{\eta}} = \boldsymbol{J}_{\Theta}(\boldsymbol{\eta})\dot{\boldsymbol{\nu}} + \dot{\boldsymbol{J}}_{\Theta}(\boldsymbol{\eta})\boldsymbol{\nu} \tag{38}$$
$$\tag{39}$$

and

$$\boldsymbol{M}^*(\boldsymbol{\eta}) = \boldsymbol{J}_{\Theta}^{-T}(\boldsymbol{\eta})\boldsymbol{M}\boldsymbol{J}_{\Theta}^{-1}(\boldsymbol{\eta}) \tag{40}$$
$$\boldsymbol{C}^*(\boldsymbol{\nu},\boldsymbol{\eta}) = \boldsymbol{J}_{\Theta}^{-T}(\boldsymbol{\eta})[\boldsymbol{C}(\boldsymbol{\nu}) - \boldsymbol{M}\boldsymbol{J}_{\Theta}^{-1}(\boldsymbol{\eta})\dot{\boldsymbol{J}}_{\Theta}(\boldsymbol{\eta})]\boldsymbol{J}_{\Theta}^{-1}(\boldsymbol{\eta}) \tag{41}$$
$$\boldsymbol{D}^*(\boldsymbol{\nu},\boldsymbol{\eta}) = \boldsymbol{J}_{\Theta}^{-T}(\boldsymbol{\eta})\boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{J}_{\Theta}^{-1}(\boldsymbol{\eta}) \tag{42}$$
$$\boldsymbol{g}^*(\boldsymbol{\eta}) + \boldsymbol{g}_0^*(\boldsymbol{\eta}) = \boldsymbol{J}_{\Theta}^{-T}(\boldsymbol{\eta})[\boldsymbol{g}(\boldsymbol{\eta}) + \boldsymbol{g}_0] \tag{43}$$
$$\boldsymbol{\tau}^* = \boldsymbol{J}_{\Theta}^{-T}(\boldsymbol{\eta})\boldsymbol{\tau} \tag{44}$$
$$\tag{45}$$

Another viable option is to apply MPC tracking using the full nonlinear dynamics model. This nonlinear model is a good representation of the vehicle dynamics and the derived input sequences should therefore be more accurate. The downside with this approach is that all hydrodynamic derivatives need to be determined and the calculation time for nonlinear MPC is significantly higher compared to linear MPC and might make this route infeasible with the current AUV hardware configuration.

Since the vehicle is underactuated in both sway and heave, another approach is to create a new dynamics model where underactuated states are indirectly controlled by actuated states and combine this model with better suited control objective. This new model could also be separated into a lateral and a longitudinal subsystems that are more or less decoupled. This separation would require the use of two different controllers but these subsystems are significantly less complex making a state feedback solutions more viable.

Since this thesis indicates that LQR is a viable control method for waypoint tracking, a compelling alternative is to keep improving the LQR controller and test it in the physical system.

References

[1] C. Deutsch, L. Moratelli, S. Thuné and J. Kuttenkeuler. *"Design of an AUV Research Platform for Demonstration of Novel Technologies"*. Autonomous Underwater Vehicle Workshop, Portugal, 2018, IEEE/OES.

[2] Symbolic Math Toolbox. *https://www.mathworks.com/help/symbolic/index.html?s_tid=CRUX_lftnav*. [Accessed Feb. 24 2020]

[3] Yalmip. *https://yalmip.github.io*. [Accessed Feb. 24 2020]

[4] Gurobi Optimization - The fastest solver. *https://www.gurobi.com*. [Accessed Feb. 24 2020]

[5] *Nomenclature for Treating the Motion of a Submerged Body Through a Fluid*. The Society of Naval Architects and Marine Engineers, Technical and Research Bulletin No. 1-5, April 1950, pp. 1-15.

[6] Thor I. Fossen. *"Handbook of Marine Craft Hydrodynamics And Motion Control"*. John Wiley & Sons, United Kingdom, 2011.

[7] Gimbal Angles. *https://www.hq.nasa.gov/alsj/gimbals.html* [Accessed Feb. 25 2020]

[8] Thor I. Fossen and Svein I. Sagatun. *"Adaptive Control of nonlinear underwater robotics systems"*. Modeling, Identification and Control, Vol. 12, No. 2, 1991, pp. 95-105.

[9] E. Torenbeek and H. Wittenberg. *"Flight Physics, Essentials of Aeronautical Disciplines and Technology, with Historical Notes"*. Springer, Netherlands, 2002.

[10] L. Ljung and T. Glad. *"Modeling and Identification of Dynamic Systems"*. Studentlitteratur AB, Sweden, 2016.

[11] T. Glad and L. Ljung. *"Reglerteknik, Grundläggande Teori"*. Studentlitteratur AB, Sweden, 2006.

[12] M. Johansson *"Notes on Model Predictive Control"*. Decision and control systems, Royal Institute of Technology, Sweden, 2017.

[13] A. Lindquist and J. Sand. revised by X. Hu *"An Introduction to Mathematical Systems Theory"*. Optimization and systems theory, Royal Institute of Technology, Sweden, 2019.

[14] W. Chang and C. Yu. *The relative gain for non-square multivariable systems*. Chemical Engineering Science, VOL. 45, No 5, 1990.

[15] Hassan K. Khalil. *"Nonlinear Control"*. Pearson Education Limited, England, 2015.

[16] U. Jönsson in collaboration with C. Trygger and P. Ögren. *"Optimal Control Lecture Notes"*. Optimization and systems theory, Royal Institute of Technology, Sweden, 2010.

[17] N. zmeneno. *"Introduction to Model Predictive Control - Tracking"*. Electrical Engineering, Czech Technical University, Czech, 2017.

[18] Michael A. Stephens, C Manzie and Malcolm C. Good. *"Model Predictive Control for Reference Tracking on an Industrial Machine Tool Servo Drive"*. IEEE Transactions on industrial informatics, VOL. 9, No. 2, May 2013.

[19] David D. Ruscio *"Model Predictive Control with Integral Action: A simple MPC algorithm"*. Modeling, Identification and Control, Vol. 34, No. 3, 2013, pp. 119-129.

TRITA-EECS-EX-2020:79
www.kth.se